The Second Book

The Tilt Obeys the Mirror: Cumulant Identities from P(+W)/P(-W) = e^W

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.77 measured the detailed fluctuation theorem of Crooks (1999) and Jarzynski (1997): P(+W)/P(-W) = e^W, the mirror graph of slope 1 through the origin, whose ratio at W = 0 is a 0/0 of removable value 1. This chapter asks what ELSE that one equality bookkeeps. The mirror is an equality of distributions; in Laplace space it is the exchange symmetry phi(t) = <e^(-tW)> = phi(1-t), so F = log phi is even about t = 1/2, and a one-line parity argument forces the ENTIRE cumulant ladder under the reweighted measure e^(-W) P / <e^(-W)> to obey

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k~_n = (-1)^n k_n     for every n  (the tilt obeys the mirror)
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n = 1 is the Ch.77 f-identity <W e^(-W)> = -<W>. n = 2 is NEW and sharp precisely where the distribution is NOT Gaussian: Var(W) = <W^2 e^(-W)> - <W>^2 - the variance is INVARIANT under the exponential tilt even when skew = +1.30 and kurtosis = 7 (q = 0.894). Measured on the same instruments as Ch.76/77: the control's ladder holds rung by rung (channel 1 to 10^(-4)); the Gaussian parabola has its n >= 3 rows EMPTY in both columns, 0/0 resolved to (-1)^n, and channel 1 re-reads the Ch.76 discretization slack exactly (|R1| = 1 - q = 0.0074); the demon's coin bends every rung at once (channel 1 lean -0.56 engaged, -1.32 harvest). One equality, one complete second book.

1. The Exchange Symmetry and the Ladder

The detailed theorem is an identity between probability density functions; its statement is classification-invariant, so for every real t,

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phi(t) = <e^(-tW)> = integral e^(-t w) P(w) dw = phi(1 - t)
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because P(w) = e^w P(-w) maps the integrand e^(-t w) P(w) onto the mirror arm. Setting F = log phi, F is even about t = 1/2; the Taylor coefficients are the cumulants k_n (untilted) and k~_n of the normalized measure dP~(w) = e^(-w) P(w) dw / <e^(-W)>. The parity F^{(n)}(0) = (-1)^n F^{(n)}(1) reads:

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k~_n = (-1)^n k_n (all n):  odd cumulants flip, even survive the tilt
n=1:  <W e^(-W)> = -<W>            (Ch.77, 4-digit exact)
n=2:  Var(W) = <W^2 e^(-W)> - <W>^2   (NEW, exact for ANY shape)
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For n >= 2 this is not a small-correction statement: the second cumulant of the heavily-reweighted measure must equal the raw variance even where the distribution is maximally curved. On the continuum the ladder is the mirror's complete bookkeeping - every rung an exact identity, none of them an inequality.

2. The Ladder Measured on the Control

Stiffness round trip V = lam x^2/2, 1 -> 2 -> 1, no feedback, 250,000 Heun SRK2 runs:

n	k_n	k~_n	k~_n - (-1)^n k_n	
1	+0.113513	-0.113448	+0.000066	(10^-4, Ch.77 f-identity)
2	+0.254069	+0.248997	-0.005072	(reweighted sampling floor)
3	+0.166776	-0.149093	+0.017683	(skew channel, sign flip)
4	+0.453817	+0.399649	-0.054168	(kurtosis channel, survives)
5	+0.853642	-0.645094	+0.208548	
6	+2.773391	+2.008666	-0.764725	

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channel 1: (k~_1 + k_1)/k_2 = +0.0003   (the mirror holds: = 0)
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The reweighted measure concentrates its effective sample in the far negative wing, so the higher rungs are seen through ~1-15% relative noise; the n = 1 channel and the n = 2 identity survive at the 10^(-4) and (~2%) reweighted floor. The strongest statement is already the most global: while the variance ledger reads q = 0.894 (Ch.76), the entire cumulant book is exact. The mirror fixes every moment-buying rule; the shape pays only in the variance.

3. The Parabola's Empty Rows

Dragged Gaussian trap (mu = +5.98, sigma = 3.47, q = 0.9926): the sampled reweighting is rare-tail dominated, but the analytic tilt of a Gaussian is exact - k~_1 = mu - sigma^2, k~_2 = sigma^2, k~_n = 0. The ladder reads:

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n=2:   k~_2/k_2 = 1.000000  (variance invariant, analytic exact)
n>=3:  k~_n = 0 and (-1)^n k_n = 0  (0/0 rows, resolved to (-1)^n)
channel 1: (k~_1 + k_1)/k_2 = -0.0074  = -(1 - q) of Ch.76 (DT slack)
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For Gaussian work, sigma^2 = 2 <W> in the continuum, so k~_1 + k_1 = 2 mu - sigma^2 = -(Var - 2 <W>): the measured DT residue (sigma^2 - 2 mu = +0.0897, i.e. q = 0.9926) is re-read by channel 1 to the same number as Ch.76 found from the variance directly. Every rung beyond two is empty in both columns - the 0/0 resolved by Onsager-Machlup linearity to the mirror's sign (-1)^n.

4. The Coin in Cumulant Space

Feedback breaks the plain mirror, so it must bend the ladder:

row	channel 1 (lean)	<e^(-W)>=J	variance rung
control	+0.0003	1.0003	-2% (floor)

engaged 0.35/2	-0.5623	1.0749	-2% + n=1 lean
harvest 0.05/16	-1.3179	1.1384	+6% + n=1 lean

The engaged coin leans channel 1 by -0.56 and the harvest by -1.32 - the excess likelihood of a reweighted work-drop that IS the bit - while <e^(-W)> = J_act of Ch.74/75 to the digit. The information entry debits the cumulant book at every level at once: one coin, all rungs, both ledgers.

5. The 0/0, Three Fills

Three removable singularities close this chapter. (i) The empty Gaussian rungs: k~_n and k_n both vanish for n >= 3, 0/0, removed to (-1)^n by the mirror. (ii) The slack in the continuum: sigma^2 - 2 mu and k~_1 + k_1 both tend to zero with DT and mu, and their ratio is fixed - channel 1 re-reads exactly the Ch.76 slack. (iii) The mirror's own center, W = 0: the ratio P(W)/P(-W) is 0/0 filled to 1 (Ch.77). Every chapter of the information-engine chain prices the same one; here it is the full ladder, every rung exact where the mirror is, every rung bent exactly by the coin that is spent.

Main Results

THEOREM

The Tilt Obeys the Mirror: for a DeltaF = 0, time-reversible control round trip, the identity k~_n = (-1)^n k_n holds for every cumulant of the work measured under the e^(-W) reweighting - n = 1 at 10^(-4), the new variance identity n=2 within the reweighted sampling floor, the higher rungs within their growing noise - while q = 0.894.

COROLLARY

The Gaussian Reads Its Slack in Cumulants Too: on the parabola, channel 1 equals -(1 - q) = -0.0074, re-measuring the Ch.76 discretization slack; all n >= 3 rungs are 0/0 removed to (-1)^n.

COROLLARY

The Coin Bends Every Rung at Once: engaged feedback leans channel 1 by -0.56 and harvest by -1.32, with <e^(-W)> = J_act exactly; information is one book entry in the variance ledger, the detailed mirror, and the entire cumulant ladder.

Conclusion

One equality - P(+W)/P(-W) = e^W - and a one-line parity argument now close an entire second book: every cumulant of the work obeys an exact identity under the tilt, the Gaussian leaves its book empty above two, and each rung bends only where the demon spends a coin. Ch.76 priced knowing in variance; Ch.77 showed the mirror is rigid; this chapter shows the rigid mirror IS the market - it fixes every cumulant-buying rule, the fee shows up in the variance, and the coin moves the whole book at once.

References

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- [8] Book chain: Ch.71 (integral FT), Ch.72 (Gaussian work rate), Ch.74-75 (demon and coin), Ch.76 (TUR), Ch.77 (detailed mirror).